D040: Editorial Apparatus

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Duality for Representations of a Reductive Group over a Finite Field

Bibliographic identity

Joint article by Pierre Deligne (I.H.E.S., Bures-sur-Yvette) and George Lusztig (Massachusetts Institute of Technology), published in *Journal of Algebra* **74** (1982), 284–291. The printed record states "Communicated by W. Feit" and "Received April 6, 1981." The byline is joint throughout this edition; neither author is treated as a secondary contributor.

Witness identities and authority

- **Controlling authority.** `D040_AUTHORITY.pdf`; eight pages; 349,015 bytes. The article text, formulae, diagram, pagination, and boundaries are checked against this witness.
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- **Comparator.** `D040_COMPARATOR_ONLY.pdf`; eight pages; 218,978 bytes. It is used only to locate possible discrepancies and does not override the controlling authority.
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- **Preserved prior-work bundle.** `D040_ZERO_ACCEPTED_PRIOR_WORK.zip`; 908,870 bytes. No reading is adopted merely because it occurs in this bundle; every retained reading is resolved against the authority.
  SHA-256 `CDD2A91B2447D1E6FD089A5E2DF57251A13F45774F92295193A041FF3C0BF1C9`

Language and edition relation

The article's source language is **English**. The source-language edition and the English reader therefore represent the same linguistic text, although they remain separately identified editions. Any French edition is a **translation from the English source**, not an alternative source-language witness. Translation must preserve the title, joint authorship, mathematical notation, cross-references, and the distinction between article text and editorial apparatus.

Editorial scope

The transcription regularizes page layout for an A4 reader. Clear typographical errors may be regularized only where the authority and mathematical context are decisive; substantive mathematical readings are never silently altered. Printed folios, running heads, and the first-page publisher and rights lines are not article body; the physical pagination and terminal blank remainder are recorded below. The comparator and prior-work bundle are subordinate evidence only.

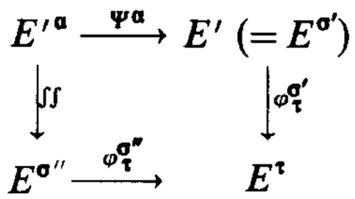
Page topology

The authority comprises eight physical PDF pages, corresponding one-for-one to printed pages 284–291.

Physical	Printed	Content boundary and principal anchors
1	284	Journal line; title; joint byline and affiliations; communication and receipt lines; section 1; notation; definition of $E_{(I)}$ begins.
2	285	Continuation of $E_{(I)}$; maps φ and $\widetilde{\varphi}$; (1.1)–(1.2); section 2 theorem and (2.1); section 3 begins.
3	286	Section 3 hypotheses (a)–(b); orbit decomposition (3.2); face maps; Lemma 3.3.
4	287	(3.4)–(3.5); simplicial-cone and reduced-chain-complex argument; section 4 and Step 1 begin.
5	288	Step 1 dimension argument; Step 2 on opposed parabolics and eigenvalues; Step 3 begins.
6	289	Step 3 module construction and commutative diagram; Steps 4–5; cone adjacency and isomorphism chain.
7	290	Step 5 continuation; Step 6; close of Lemma 3.3 and the theorem; section 5 corollary; (5.1)–(5.2); proof ends.
8	291	Complete References, items 1–4; blank lower remainder; physical end of article.

Image-fallback ledger: page 6 diagram

The Step 3 commutative diagram is retained from the controlling authority as a page-local raster derivative because the captured text layer does not preserve its node and arrow-label topology reliably. The fallback is not a reconstructed mathematical diagram.



Source address	Authority physical page 6 / printed page 289, Step 3
Derivative	`D040_p006_diagram_authority_400dpi.png`; 740 × 360 pixels; 9,939 bytes
SHA-256	`B932DC7CB4ADB6790BD47C7F731914E38AC45426C7248A3F4490BD4D2DE4285F`
Disposition	Reproduced as an authority crop; no node, label, or arrow is editorially inferred.

Source-checked correction ledger

These entries identify readings for which comparison exposed a discrepancy or layout risk. Each disposition follows the controlling authority.

ID	Address	Authority-based disposition
C001	p. 284, opening	Retain "At level of characters" and the source punctuation around the parenthetical example; do not modernize the phrase.
C002	p. 284, notation	Retain the bar on $\bar{S}$, the set of Frobenius orbits; do not substitute a tilde.
C003	p. 285, before (1.1)	Retain $U_P \supset U_{P'}$, hence $E^{U_P} \subset E^{U_{P'}}$.
C004	p. 285, (1.2)	Retain the displayed complex with arrow labels d_0, d_1, and d_2.
C005	p. 285, section 2	Retain the unnumbered heading **THEOREM.**; do not introduce a new theorem counter.
C006	p. 286, (3.2)	Keep U_Q-invariants outside each constrained inner direct sum, and retain the vanishing condition involving $U_Q \cap P$.
C007	pp. 286–287	Retain the face map $\delta^J_{J'}$ and orbit-indexed maps $\Psi^\sigma_{\sigma'}$; do not collapse them to subset-indexed maps.
C008	p. 287, (3.4)–(3.5)	Terms increase from $\lvert J\rvert = \lvert I_0\rvert$ to $\lvert J\rvert = \lvert I_0\rvert + 1$; retain the initial zero terms, the two distinct unions, and terminal homology $\cong K$.
C009	p. 288, Step 1	Retain the complete outer and inner sums, $\left(\bigoplus_{P\in B} E_P\right)^{U_Q}$, and the count $\#(\mathcal{P}_{\tilde{J}})$.
C010	p. 289, Step 3	Retain $R' \in \mathcal{P}_{J'}$, $(R', P) \in \sigma'$, and the distinct maps in the commutative argument; use the authority crop for the diagram.
C011	pp. 289–290	Retain the orbit-indexed compositions through Steps 4–6; no undefined subset index is introduced.
C012	p. 290, section 5	Retain the unnumbered heading **COROLLARY (see Alvis [1]).** and the second relation $E'^{\#} \cong E'$ in (5.1).
C013	p. 290, (5.2)	Retain $\langle E_i, E_j^{\#}\rangle = \langle E_i^{\#}, E_j\rangle$ and the U_{P_I}-invariant-space inner product; the next sum pairs $E_i^{\#}$ with E_j, not $E_j^{\#}$.
C014	p. 290, conclusion	Retain $\pi^2(i)$ in the terminal pairing and conclusion; do not replace it by $\pi^{-1}(i)$.
C015	p. 291	Preserve all four references and record the blank lower remainder and physical end, even when the A4 reader reflows the list.

Editorial note

The ledger records source readings; it does not claim to correct the authors' mathematics. Equation labels, result headings, notation, and terminal boundaries are preserved where the authority is decisive. The English transcription is an editorial reading of the authority, while the authority PDF remains the final arbiter in any future dispute.